The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Germ-Anchored Exact-Shell Transport-Section Core
from Exact-Shell-Transport-Section-Generating
Substrate Proto-Section Dynamics

Abstract

The current primitive-reservoir observer-reduction frontier already sharpened the live burden inside the proto-germ line: the larger germ-anchored exact-shell transport core was shown to collapse benchmark-facingly to the smaller germ-anchored exact-shell transport-section core $\mathfrak{P}^{\mathrm{sub,proto,sec}}$. Under the routing safeguard against recursive same-output depth drift, however, the honest next move is no longer another deeper relay that merely preserves that same section core. The next admissible benchmark-facing gain is to derive or justify the section core itself from a sharper substrate package, or else prove a still smaller theorem strictly inside it.

The present note takes the direct derivation route. For each sector it introduces an *exact-shell-transport-section-generating substrate proto-section dynamics package*: a compact convex proto-section body over the already recorded proto-germ body, an affine projection to that fixed proto body, a nonnegative smooth fiber-selection functional with unique minimizers on fibers above the fixed current germ body, and an exact proto-section shell whose projected minimizers above the fixed current germ shell reproduce the recorded current proto shell. The current germ-anchored exact-shell transport-section core is then no longer an unexplained primitive stopping point. It is the canonical projected exact-shell minimizer section selected by that deeper fiberwise substrate law.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. The positive result is therefore narrow but benchmark facing: the live burden is no longer the current section core as such, but the deeper exact-shell-transport-section-generating substrate proto-section dynamics package that canonically generates it.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has again reached a point where a smaller benchmark-facing datum has already been isolated. The current active-primary theorem showed that, once the recorded current germ nucleus is fixed, the larger germ-anchored exact-shell transport core is not the honest remaining burden. Its separate shell presentation, restricted shell projection, and ambient off-shell proto presentation collapse to the smaller germ-anchored exact-shell transport-section core $\mathfrak{P}^{\mathrm{sub,proto,sec}}$.

That compression matters because it blocks a familiar escape route. Another deeper-origin note that merely preserves the same current section core would not count as new front-line progress. The present burden is sharper: either derive or justify that section core itself from still more primitive

substrate structure, or prove a genuinely smaller theorem-level collapse, sharper necessity theorem, or sharper uniqueness theorem strictly inside it.

The present note takes the direct derivation route. Its positive claim is not that completed substrate microphysics has now been identified, and it is not that the benchmark exact-relativistic package has been altered. Its claim is narrower and benchmark facing. The current section core is shown to arise canonically as the projected exact-shell minimizer selected by a deeper proto-section package lying over the recorded proto-germ body and its fixed shell projection to the current germ body.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not introduce a second operative metric, it does not enlarge the low-energy matter law, it does not change the ordinary matter route, and it does not reopen older seed, root, foundation, ground, origin, precursor, lift, shell-restriction, selector-law, combined-response, ambient-package, or exact-shell-affine-nucleus layers as if they were again the live burden. Its narrower positive role is to show that the current section core itself is canonically generated by a deeper fiberwise substrate selector law.

2. Fixed Inputs

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}},$$

a visible reduction map

$$\mathcal{R}_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \to X_\sigma,$$

and shell-level current and equilibrium carriers

$$\mathfrak{C}_\sigma^{\mathrm{cur}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{cur}}, \qquad \mathfrak{C}_\sigma^{\mathrm{eq}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{eq}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^{\sharp}, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T^{\mathrm{matter}}_{\mu\nu}[g^{\sharp}, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$.

Definition 2 (Recorded downstream seed package)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *The active line already fixes the downstream seed package at current benchmark-facing scope:*

1. *a finite-dimensional seed state space* $\mathcal{S}^{\mathrm{seed}}_{\sigma}$, *a compact convex seed body* $\mathcal{B}^{\mathrm{seed}}_{\sigma} \subset \mathcal{S}^{\mathrm{seed}}_{\sigma}$, *the exact seed shell* $\mathcal{Z}^{\mathrm{seed}}_{\sigma} \subset \mathcal{B}^{\mathrm{seed}}_{\sigma}$, *and the exact seed support shell* $\mathcal{U}^{\mathrm{seed,sup}}_{\sigma} \subset \mathcal{Z}^{\mathrm{seed}}_{\sigma}$;

2. *a seed zero core* $\mathcal{Z}^{0,\mathrm{seed}}_{\sigma} \subset \mathcal{U}^{\mathrm{seed,sup}}_{\sigma}$, *the same hidden fiber space* $\mathcal{H}^{\mathrm{hid}}_{\sigma}$ *with compact hidden body* $D^{\mathrm{hid}}_{\sigma} \subset \mathcal{H}^{\mathrm{hid}}_{\sigma}$, *and the fixed seed split*

$$\Phi^{\mathrm{seed}}_{\sigma} : \mathcal{Z}^{0,\mathrm{seed}}_{\sigma} \times D^{\mathrm{hid}}_{\sigma} \xrightarrow{\cong} \mathcal{U}^{\mathrm{seed,sup}}_{\sigma};$$

3. *the seed hidden charge*

$$\mathcal{Q}^{\mathrm{seed}}_{\sigma} : \mathcal{H}^{\mathrm{hid}}_{\sigma} \to Q^{\mathrm{eqn}}_{\sigma},$$

a smooth observer-compatible exact seed zero-response realization

$$\zeta^{\mathrm{seed}}_{\sigma} : \mathcal{I}^{\mathrm{for}}_{\sigma} \hookrightarrow \mathcal{Z}^{0,\mathrm{seed}}_{\sigma},$$

and the fixed-data solvability / coercive package recorded through $\widetilde{K}^{0,\mathrm{seed}}_{\sigma}$ *and* $P^{\mathrm{seed,co}}_{\sigma}$;

4. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

Definition 3 (Recorded current minimal germ zero-shell transport nucleus)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *The recorded current minimal germ zero-shell transport nucleus* $\mathfrak{G}^{\mathrm{sub,germ,zstn}}$ *for sector* σ *consists of:*

1. *the germ state space* $\mathcal{G}^{\mathrm{germ}}_{\sigma}$, *compact convex germ body* $\mathcal{B}^{\mathrm{germ}}_{\sigma} \subset \mathcal{G}^{\mathrm{germ}}_{\sigma}$, *projection*

$$\Pi^{\mathrm{germ}}_{\sigma} : \mathcal{G}^{\mathrm{germ}}_{\sigma} \to \mathcal{S}^{\mathrm{seed}}_{\sigma},$$

and seed section

$$\sigma^{\mathrm{germ}\leftarrow\mathrm{seed}}_{\sigma} : \mathcal{B}^{\mathrm{seed}}_{\sigma} \to \mathcal{B}^{\mathrm{germ}}_{\sigma}$$

satisfying

$$\Pi^{\mathrm{germ}}_{\sigma}(\mathcal{B}^{\mathrm{germ}}_{\sigma}) = \mathcal{B}^{\mathrm{seed}}_{\sigma}, \qquad \Pi^{\mathrm{germ}}_{\sigma} \circ \sigma^{\mathrm{germ}\leftarrow\mathrm{seed}}_{\sigma} = \mathrm{id}_{\mathcal{B}^{\mathrm{seed}}_{\sigma}};$$

2. *the restriction section*

$$\Gamma^{\mathrm{germ}}_{\sigma} : \mathcal{G}^{\mathrm{germ}}_{\sigma} \to E^{\mathrm{sub}}_{\sigma},$$

and the exact germ shell

$$\mathcal{Z}^{\mathrm{germ}}_{\sigma} \equiv \left(\Gamma^{\mathrm{germ}}_{\sigma}\right)^{-1}(0) \cap \mathcal{B}^{\mathrm{germ}}_{\sigma},$$

with

$$\Pi_\sigma^{\mathrm{germ}}\big|_{\mathcal{Z}_\sigma^{\mathrm{germ}}} : \mathcal{Z}_\sigma^{\mathrm{germ}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{seed}};$$

3. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

By the already-recorded current frontier theorem, the support shell, zero core, hidden split, hidden charge, exact zero-response realization, and fixed-data solvability / coercive package are then canonically induced downstream from this nucleus together with the fixed seed package of Definition 2, rather than taken as additional independent benchmark-facing data.

Definition 4 (Recorded current germ-anchored exact-shell transport core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The recorded current germ-anchored exact-shell transport core* $\mathfrak{P}^{\mathrm{sub,proto,xtc}}$ *for sector* σ *consists of:*

1. *the proto-germ state space* $\mathcal{P}_\sigma^{\mathrm{proto}}$*, compact convex proto-germ body* $\mathcal{B}_\sigma^{\mathrm{proto}} \subset \mathcal{P}_\sigma^{\mathrm{proto}}$*, and affine surjection*

$$\Pi_\sigma^{\mathrm{proto}} : \mathcal{P}_\sigma^{\mathrm{proto}} \to \mathcal{G}_\sigma^{\mathrm{germ}}$$

satisfying

$$\Pi_\sigma^{\mathrm{proto}}(\mathcal{B}_\sigma^{\mathrm{proto}}) = \mathcal{B}_\sigma^{\mathrm{germ}};$$

2. *the restriction section*

$$\Gamma_\sigma^{\mathrm{proto}} : \mathcal{P}_\sigma^{\mathrm{proto}} \to E_\sigma^{\mathrm{sub}},$$

the exact proto-germ shell

$$\mathcal{Z}_\sigma^{\mathrm{proto}} \equiv \left(\Gamma_\sigma^{\mathrm{proto}}\right)^{-1}(0) \cap \mathcal{B}_\sigma^{\mathrm{proto}},$$

and the exact-shell transport diffeomorphism

$$\Pi_\sigma^{\mathrm{proto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{proto}}} : \mathcal{Z}_\sigma^{\mathrm{proto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{germ}};$$

3. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

Definition 5 (Recorded current germ-anchored exact-shell transport-section core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ *and the recorded transport core of Definition 4. The recorded current germ-anchored exact-shell transport-section core* $\mathfrak{P}^{\mathrm{sub,proto,sec}}$ *for sector* σ *consists of:*

1. *the canonical exact-shell transport section*

$$\mathfrak{s}_\sigma^{\mathrm{proto,sec}} \equiv \left(\Pi_\sigma^{\mathrm{proto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{proto}}}\right)^{-1} : \mathcal{Z}_\sigma^{\mathrm{germ}} \to \mathcal{Z}_\sigma^{\mathrm{proto}} \subset \mathcal{B}_\sigma^{\mathrm{proto}} \subset \mathcal{P}_\sigma^{\mathrm{proto}};$$

2. *benchmark faithfulness in the sense of Definition 4.*

The exact proto shell and the restricted shell projection back to $\mathcal{Z}_\sigma^{\mathrm{germ}}$ *are understood as derived data rather than as separately primitive section-core data.*

Remark 1. *Definitions 1–5 are fixed inputs. The primitive continuation data, the one operative metric, the ordinary matter route, the downstream seed package, the recorded current germ nucleus,*

and the recorded current transport-core package are not reopened here as primitive stopping points. The present note asks only why the current section core itself should arise from a sharper substrate package.

3. Exact-Shell-Transport-Section-Generating Substrate Proto-Section Dynamics

Definition 6 (Exact-shell-transport-section-generating substrate proto-section dynamics)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *An* exact-shell-transport-section-generating substrate proto-section dynamics package $\mathfrak{Q}^{\mathrm{sub,proto,secgen}}$ *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean proto-section state space* $\mathcal{Q}_\sigma^{\mathrm{psec}}$*, a compact convex proto-section body*
$$\mathcal{B}_\sigma^{\mathrm{psec}} \subset \mathcal{Q}_\sigma^{\mathrm{psec}}$$
with nonempty interior, and an affine surjection
$$\Pi_\sigma^{\mathrm{psec}} : \mathcal{Q}_\sigma^{\mathrm{psec}} \to \mathcal{P}_\sigma^{\mathrm{proto}}$$
satisfying
$$\Pi_\sigma^{\mathrm{psec}}(\mathcal{B}_\sigma^{\mathrm{psec}}) = \mathcal{B}_\sigma^{\mathrm{proto}}.$$
For each $x \in \mathcal{B}_\sigma^{\mathrm{germ}}$*, write*
$$\mathcal{F}_\sigma^{\mathrm{psec}}(x) \equiv \left\{ y \in \mathcal{B}_\sigma^{\mathrm{psec}} : \Pi_\sigma^{\mathrm{proto}}(\Pi_\sigma^{\mathrm{psec}}(y)) = x \right\}.$$
Each such fiber is required to be nonempty, compact, and convex.

2. *A nonnegative smooth fiber-selection functional*
$$\mathscr{E}_\sigma^{\mathrm{psec}} : \mathcal{Q}_\sigma^{\mathrm{psec}} \to [0, \infty)$$
such that, for every $x \in \mathcal{B}_\sigma^{\mathrm{germ}}$*, the restriction of* $\mathscr{E}_\sigma^{\mathrm{psec}}$ *to* $\mathcal{F}_\sigma^{\mathrm{psec}}(x)$ *is strictly convex and attains a unique minimizer. The resulting minimizer depends smoothly on* x*.*

3. *A closed embedded exact proto-section shell*
$$\mathcal{Z}_\sigma^{\mathrm{psec}} \subset \mathcal{B}_\sigma^{\mathrm{psec}}$$
satisfying
$$\mathcal{Z}_\sigma^{\mathrm{psec}} = \left\{ y \in \mathcal{B}_\sigma^{\mathrm{psec}} : \Pi_\sigma^{\mathrm{proto}}(\Pi_\sigma^{\mathrm{psec}}(y)) \in \mathcal{Z}_\sigma^{\mathrm{germ}},\ \mathscr{E}_\sigma^{\mathrm{psec}}(y) = \min_{z \in \mathcal{F}_\sigma^{\mathrm{psec}}(\Pi_\sigma^{\mathrm{proto}}(\Pi_\sigma^{\mathrm{psec}}(y)))} \mathscr{E}_\sigma^{\mathrm{psec}}(z) \right\},$$
and
$$\Pi_\sigma^{\mathrm{psec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{psec}}} : \mathcal{Z}_\sigma^{\mathrm{psec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{proto}}$$
is a diffeomorphism.

4. *Benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

Remark 2. *Definition 6 is intentionally written over the already fixed proto-germ body and its shell projection to the recorded current germ body. It does not restore the larger transport-core presentation as the live burden. Its positive role is narrower: the current section core is generated as the projected exact-shell minimizer selected by a deeper fiberwise substrate law.*

4. Induced Current Section Core

Definition 7 (Canonical proto-section minimizer and induced projected transport)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ *and a benchmark-faithful proto-section dynamics package in the sense of Definition 6. Define the canonical proto-section minimizer by*

$$\mathfrak{m}_\sigma^{\mathrm{psec}}(x) \equiv \operatorname*{arg\,min}_{y \in \mathcal{F}_\sigma^{\mathrm{psec}}(x)} \mathscr{E}_\sigma^{\mathrm{psec}}(y), \qquad x \in \mathcal{B}_\sigma^{\mathrm{germ}},$$

and define the induced projected exact-shell transport section by

$$\widehat{\mathfrak{s}}_\sigma^{\mathrm{proto,sec}}(x) \equiv \Pi_\sigma^{\mathrm{psec}}\big(\mathfrak{m}_\sigma^{\mathrm{psec}}(x)\big), \qquad x \in \mathcal{Z}_\sigma^{\mathrm{germ}}.$$

Remark 3. *Definition 7 supplies a deeper minimizer model for the currently live benchmark-facing datum. The present theorem will show that the induced projected exact-shell transport is exactly the recorded current section core* $\mathfrak{s}_\sigma^{\mathrm{proto,sec}}$*.*

5. Canonical Recovery of the Current Section Core

Proposition 1 (The canonical proto-section minimizer is well defined)**.** *For every benchmark-faithful proto-section dynamics package, the map* $\mathfrak{m}_\sigma^{\mathrm{psec}}$ *of Definition 7 is a smooth section of the composite projection* $\Pi_\sigma^{\mathrm{proto}} \circ \Pi_\sigma^{\mathrm{psec}}$*, so that*

$$\Pi_\sigma^{\mathrm{proto}} \circ \Pi_\sigma^{\mathrm{psec}} \circ \mathfrak{m}_\sigma^{\mathrm{psec}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{germ}}}.$$

Proof. Definition 6(1) makes each fiber $\mathcal{F}_\sigma^{\mathrm{psec}}(x)$ nonempty, compact, and convex. Definition 6(2) states that $\mathscr{E}_\sigma^{\mathrm{psec}}$ has a unique minimizer on each such fiber and that the minimizer depends smoothly on x. Definition 7 therefore gives a well-defined smooth map

$$\mathfrak{m}_\sigma^{\mathrm{psec}} : \mathcal{B}_\sigma^{\mathrm{germ}} \to \mathcal{B}_\sigma^{\mathrm{psec}}.$$

Because every minimizer lies in the fiber over the point at which it is selected, we have

$$\Pi_\sigma^{\mathrm{proto}}\Big(\Pi_\sigma^{\mathrm{psec}}\big(\mathfrak{m}_\sigma^{\mathrm{psec}}(x)\big)\Big) = x$$

for every $x \in \mathcal{B}_\sigma^{\mathrm{germ}}$, which is the displayed identity. □

Proposition 2 (The exact proto-section shell is the shell restriction of the canonical minimizer)**.** *For every benchmark-faithful proto-section dynamics package,*

$$\mathcal{Z}_\sigma^{\mathrm{psec}} = \mathfrak{m}_\sigma^{\mathrm{psec}}\big(\mathcal{Z}_\sigma^{\mathrm{germ}}\big),$$

and the shell restriction

$$\mathfrak{m}_\sigma^{\mathrm{psec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{germ}}} : \mathcal{Z}_\sigma^{\mathrm{germ}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{psec}}$$

is a diffeomorphism with inverse

$$\Pi_\sigma^{\mathrm{proto}} \circ \Pi_\sigma^{\mathrm{psec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{psec}}} : \mathcal{Z}_\sigma^{\mathrm{psec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{germ}}.$$

Proof. Let $x \in \mathcal{Z}_\sigma^{\mathrm{germ}}$. By Definition 7, $\mathfrak{m}_\sigma^{\mathrm{psec}}(x)$ is the unique minimizer of $\mathscr{E}_\sigma^{\mathrm{psec}}$ on the fiber $\mathcal{F}_\sigma^{\mathrm{psec}}(x)$. Since $x \in \mathcal{Z}_\sigma^{\mathrm{germ}}$, Definition 6(3) implies

$$\mathfrak{m}_\sigma^{\mathrm{psec}}(x) \in \mathcal{Z}_\sigma^{\mathrm{psec}}.$$

Hence

$$\mathfrak{m}_\sigma^{\mathrm{psec}}\big(\mathcal{Z}_\sigma^{\mathrm{germ}}\big) \subset \mathcal{Z}_\sigma^{\mathrm{psec}}.$$

Conversely, if $y \in \mathcal{Z}_\sigma^{\mathrm{psec}}$, then

$$x \equiv \Pi_\sigma^{\mathrm{proto}}\big(\Pi_\sigma^{\mathrm{psec}}(y)\big)$$

lies in $\mathcal{Z}_\sigma^{\mathrm{germ}}$, and Definition 6(3) says that y is a minimizer of $\mathscr{E}_\sigma^{\mathrm{psec}}$ on $\mathcal{F}_\sigma^{\mathrm{psec}}(x)$. By uniqueness of that minimizer, $y = \mathfrak{m}_\sigma^{\mathrm{psec}}(x)$. This proves the displayed equality.

Definition 6(3) gives a diffeomorphism

$$\Pi_\sigma^{\mathrm{psec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{psec}}} : \mathcal{Z}_\sigma^{\mathrm{psec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{proto}}.$$

Definition 4(2) gives a diffeomorphism

$$\Pi_\sigma^{\mathrm{proto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{proto}}} : \mathcal{Z}_\sigma^{\mathrm{proto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{germ}}.$$

Their composition is therefore a diffeomorphism

$$\Pi_\sigma^{\mathrm{proto}} \circ \Pi_\sigma^{\mathrm{psec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{psec}}} : \mathcal{Z}_\sigma^{\mathrm{psec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{germ}},$$

and the displayed equality shows that its inverse is exactly $\mathfrak{m}_\sigma^{\mathrm{psec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{germ}}}$. □

Proposition 3 (The projected shell transport reproduces the current section core)**.** *For every benchmark-faithful proto-section dynamics package,*

$$\widehat{\mathfrak{s}}_\sigma^{\mathrm{proto,sec}} = \Big(\Pi_\sigma^{\mathrm{proto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{proto}}}\Big)^{-1} = \mathfrak{s}_\sigma^{\mathrm{proto,sec}} \qquad \textit{on } \mathcal{Z}_\sigma^{\mathrm{germ}}.$$

Proof. Fix $x \in \mathcal{Z}_\sigma^{\mathrm{germ}}$. By Definition 7,

$$\widehat{\mathfrak{s}}_\sigma^{\mathrm{proto,sec}}(x) = \Pi_\sigma^{\mathrm{psec}}\big(\mathfrak{m}_\sigma^{\mathrm{psec}}(x)\big).$$

By Proposition 2, $\mathfrak{m}_\sigma^{\mathrm{psec}}(x) \in \mathcal{Z}_\sigma^{\mathrm{psec}}$, so the displayed point lies in $\mathcal{Z}_\sigma^{\mathrm{proto}}$. Applying $\Pi_\sigma^{\mathrm{proto}}$ gives

$$\Pi_\sigma^{\mathrm{proto}}\big(\widehat{\mathfrak{s}}_\sigma^{\mathrm{proto,sec}}(x)\big) = \Pi_\sigma^{\mathrm{proto}}\Big(\Pi_\sigma^{\mathrm{psec}}\big(\mathfrak{m}_\sigma^{\mathrm{psec}}(x)\big)\Big) = x$$

by Proposition 1. Therefore $\widehat{\mathfrak{s}}_\sigma^{\mathrm{proto,sec}}(x)$ is a point of $\mathcal{Z}_\sigma^{\mathrm{proto}}$ projecting to $x \in \mathcal{Z}_\sigma^{\mathrm{germ}}$. Since

$$\Pi_\sigma^{\mathrm{proto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{proto}}} : \mathcal{Z}_\sigma^{\mathrm{proto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{germ}}$$

is a diffeomorphism by Definition 4(2), that point is uniquely determined and equals

$$\left(\Pi_\sigma^{\mathrm{proto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{proto}}}\right)^{-1}(x) = \mathfrak{s}_\sigma^{\mathrm{proto,sec}}(x).$$

□

Proposition 4 (The projected shell transport is canonical)**.** *Fix a benchmark-faithful proto-section dynamics package. If*

$$s : \mathcal{Z}_\sigma^{\mathrm{germ}} \to \mathcal{B}_\sigma^{\mathrm{psec}}$$

is any smooth map satisfying

$$s(x) \in \mathcal{F}_\sigma^{\mathrm{psec}}(x), \qquad \mathscr{E}_\sigma^{\mathrm{psec}}(s(x)) = \min_{y \in \mathcal{F}_\sigma^{\mathrm{psec}}(x)} \mathscr{E}_\sigma^{\mathrm{psec}}(y)$$

for every $x \in \mathcal{Z}_\sigma^{\mathrm{germ}}$, then

$$s = \mathfrak{m}_\sigma^{\mathrm{psec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{germ}}}$$

and

$$\Pi_\sigma^{\mathrm{psec}} \circ s = \widehat{\mathfrak{s}}_\sigma^{\mathrm{proto,sec}} = \mathfrak{s}_\sigma^{\mathrm{proto,sec}}.$$

Proof. For each $x \in \mathcal{Z}_\sigma^{\mathrm{germ}}$, Definition 6(2) supplies a unique minimizer of $\mathscr{E}_\sigma^{\mathrm{psec}}$ on $\mathcal{F}_\sigma^{\mathrm{psec}}(x)$. Therefore the assumed map s must agree pointwise with the unique minimizer $\mathfrak{m}_\sigma^{\mathrm{psec}}(x)$. The projected identity then follows from Proposition 3. □

Proposition 5 (Composite transport to the fixed seed shell factors through the induced section)**.** *For every benchmark-faithful proto-section dynamics package, the composite shell transport to the fixed seed shell is reconstructed from the induced projected section by*

$$\Pi_\sigma^{\mathrm{germ}}\big|_{\mathcal{Z}_\sigma^{\mathrm{germ}}} \circ \left(\widehat{\mathfrak{s}}_\sigma^{\mathrm{proto,sec}}\right)^{-1} : \mathcal{Z}_\sigma^{\mathrm{proto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{seed}}.$$

Proof. Proposition 3 gives

$$\left(\widehat{\mathfrak{s}}_\sigma^{\mathrm{proto,sec}}\right)^{-1} = \Pi_\sigma^{\mathrm{proto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{proto}}} : \mathcal{Z}_\sigma^{\mathrm{proto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{germ}}.$$

Definition 3(2) gives the fixed shell diffeomorphism

$$\Pi_\sigma^{\mathrm{germ}}\big|_{\mathcal{Z}_\sigma^{\mathrm{germ}}} : \mathcal{Z}_\sigma^{\mathrm{germ}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{seed}}.$$

Composing these two diffeomorphisms yields the displayed transport from $\mathcal{Z}_\sigma^{\mathrm{proto}}$ to $\mathcal{Z}_\sigma^{\mathrm{seed}}$. □

Proposition 6 (All current benchmark-facing uses of the section core factor through the deeper package)**.** *For every benchmark-faithful proto-section dynamics package, all current benchmark-facing uses of the germ-anchored exact-shell transport-section core on the active line already factor through the deeper proto-section package together with the fixed current germ nucleus $\mathfrak{G}^{\mathrm{sub,germ,zstn}}$. In particular, no additional primitive benchmark-facing datum beyond the deeper package, the fixed*

current germ nucleus, and benchmark faithfulness is required to recover the present section-core consequences.

Proof. Proposition 3 reconstructs the recorded current section core $\mathfrak{s}_\sigma^{\mathrm{proto,sec}}$ from the deeper proto-section package. Proposition 5 reconstructs the composite shell transport to the fixed seed shell. The recorded current germ nucleus of Definition 3 already fixes the germ body, the seed anchor, the germ restriction section, and the exact germ shell at benchmark-facing scope; those data are not altered by replacing the present section core with its deeper proto-section source.

The previous active-primary theorem already proved that all current benchmark-facing uses of the larger transport core factor through the section core together with that fixed current germ nucleus. Since Proposition 3 recovers the same section core from the deeper proto-section package, every current benchmark-facing use of the section core and of the older larger transport-core presentation now factors through the deeper proto-section package together with the fixed current germ nucleus. □

Proposition 7 (The benchmark package remains fixed)**.** *Every benchmark-faithful proto-section dynamics package preserves the same benchmark one-metric package* (1)*. In particular, the operative metric remains exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

Proof. This is part of benchmark faithfulness in Definition 6(4). The present note only replaces the current section core by a deeper generating package. It does not alter the benchmark exact-closure package. □

6. Main Theorem

Theorem 1 (Germ-anchored exact-shell transport-section core from exact-shell-transport-section–generating substrate proto-section dynamics)**.** *Fix the primitive continuation data of Definition 1, the recorded downstream seed package of Definition 2, the recorded current germ nucleus of Definition 3, the recorded current transport core of Definition 4, and a benchmark-faithful exact-shell-transport-section-generating substrate proto-section dynamics package in the sense of Definition 6. Then, for each sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$, the current germ-anchored exact-shell transport-section core is no longer an unexplained primitive stopping point on the active line: it is recovered canonically as the projected exact-shell minimizer section of the deeper proto-section dynamics. More precisely:*

1. *the canonical proto-section minimizer $\mathfrak{m}_\sigma^{\mathrm{psec}}$ is well defined over the fixed current germ body by Proposition 1;*
2. *the exact proto-section shell is the shell restriction of that minimizer and projects diffeomorphically to the recorded current proto shell by Proposition 2;*
3. *the projected shell transport is exactly the recorded current section core $\mathfrak{s}_\sigma^{\mathrm{proto,sec}}$, and it is uniquely forced by the deeper package on $\mathcal{Z}_\sigma^{\mathrm{germ}}$, by Propositions 3 and 4;*
4. *all current benchmark-facing uses of the section core and the earlier larger transport-core presentation already factor through the deeper proto-section package together with the fixed current germ nucleus by Proposition 6;*
5. *the benchmark boundary remains fixed by Proposition 7;*

6. *consequently the remaining honest burden below the previous transport-section-core frontier is no longer the current germ-anchored exact-shell transport-section core* $\mathfrak{P}^{\mathrm{sub,proto,sec}}$ *as such. What remains is to derive or justify the deeper exact-shell-transport-section-generating substrate proto-section dynamics* $\mathfrak{Q}^{\mathrm{sub,proto,secgen}}$ *from still more primitive substrate structure, or else prove a still smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside that deeper package.*

Proof. Item (1) is Proposition 1. Item (2) is Proposition 2. Item (3) is the combined content of Propositions 3 and 4. Item (4) is Proposition 6. Item (5) is Proposition 7. Item (6) is the routing consequence of the previous items together with the active safeguard against counting another same-output deeper-origin relay as new front-line progress once the live benchmark-relevant datum has already been canonically generated. □

Corollary 1 (What must be done next)**.** *After Theorem 1, the next honest benchmark-facing manuscript must either:*

1. *derive or justify the exact-shell-transport-section-generating substrate proto-section dynamics* $\mathfrak{Q}^{\mathrm{sub,proto,secgen}}$ *from still more primitive substrate structure in a way that changes benchmark-facing status;*

2. *identify a genuinely smaller theorem-level observer-relevant datum strictly inside* $\mathfrak{Q}^{\mathrm{sub,proto,secgen}}$ *and prove a sharper collapse, sharper necessity result, or sharper uniqueness result there;*

3. *do either of the previous two while preserving the same benchmark one-metric package, the same ordinary matter route, and the same claim boundary.*

Another same-output deeper-origin relay that merely preserves this deeper proto-section package, another restatement of the current section core as if it were still independently primitive, another re-expansion back to the larger transport-core presentation, another replay of the current minimal germ zero-shell transport nucleus, another reopening of older seed, root, foundation, ground, origin, precursor, lift, shell-restriction, selector-law, combined-response, ambient-package, or exact-shell-affine-nucleus layers, or any move that introduces a second operative metric, enlarges the ordinary matter law, changes the ordinary matter route, or uses stronger promotion language does not satisfy the present burden.

Proof. Theorem 1 shows that the current section core is no longer primitive at the live frontier. Therefore a subsequent manuscript must improve on the deeper proto-section dynamics rather than merely restating the already-generated section core or re-expanding to the larger transport-core presentation. □

7. Conclusion

The positive result of this note is a disciplined derivational gain below the current proto-germ transport-section-core frontier. The previous active-primary theorem had already shown that the larger germ-anchored exact-shell transport core was not the minimal benchmark-facing datum, because its live burden collapsed to the smaller section core. The present theorem then removes that new stopping point itself: the current section core is shown to be the canonical projected exact-shell minimizer selected by a deeper proto-section package.

What survives as the live burden is therefore no longer the current section core. It is the deeper exact-shell-transport-section-generating substrate proto-section dynamics that generate that section. Once those deeper dynamics are fixed, the current transport section is uniquely forced on the fixed current germ shell, the current proto shell is recovered as the projected shell of the deeper package, and all current benchmark-facing uses of the section core and of the older larger transport-core presentation follow unchanged together with the already fixed current germ nucleus.

The scope remains disciplined. The benchmark package is unchanged: the one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. This note does not yet complete a first-principles derivation of GR from substrate microphysics, but it does remove the current section core as the live primitive stopping point and move the burden one honest level deeper on the active line.